

Abhinaba Dutta

🎓 M.Sc. Candidate in Data, Algorithms and Machine Intelligence

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Education

MSc, Data, Algorithms and Machine Intelligence

2024 – 2027 (expected)

University of Palermo, Italy

Relevant coursework: Big Data, Probabilistic Algorithms, AI & Deep Learning, Knowledge Representation

Bachelor of Technology

2017 – 2020

Maulana Abul Kalam Azad University of Technology (MAKAUT), Kolkata, India

Skills

Programming & Data: Python, SQL, JavaScript, TypeScript, Pandas, NumPy, Apache Spark

Machine Learning & NLP: Scikit-learn, PyTorch, TF-IDF, LLMs, Prompt Engineering, Model Evaluation

Biomedical & Tools: SNOMED CT, Biomedical NLP, Knowledge Graphs, MongoDB, MySQL, Git, Linux, Docker

Experience

ICAR-CNR (Italy)

03/2026 - 07/2026

Academic Intern (NLP)

Python, Scikit-learn, TF-IDF, Rule-based NLP, JSON, LLMs

- Built an offline NLP-based advisor that answers wind-turbine maintenance, inspection, safety, and troubleshooting questions from technical manuals.
- Designed a structured JSON knowledge base covering 15 turbine components and implemented rule-based intent and component detection.
- Applied TF-IDF and cosine similarity to retrieve evidence from the maintenance manual and integrated Groq/Llama as an optional answer-rewriting layer.

Ecera System Private Limited (India)

08/2023 - 05/2024

Full stack Developer

Node.js, Express.js, ReactJS, TypeScript, GIT, Cloud Deployment

- Built and deployed a full-stack LMS with Razorpay payments and Google OAuth.
- Developed gamification, authentication, and refund workflows to improve user engagement and payment handling.
- Created a responsive React and Tailwind CSS interface for desktop and mobile users.

Projects

Explainable Patient–Clinical Trial Matching

Master's Thesis Research

Python, LLMs, SNOMED CT, Biomedical NLP, Knowledge Graphs

- Developed a hybrid pipeline for matching patient descriptions with clinical-trial eligibility criteria.
- Used large language models to extract structured patient information and analyse inclusion and exclusion criteria.
- Combined medical concept normalization, biomedical semantic matching, rule-based validation, and weighted eligibility scoring.
- Generated criterion-level explanations and evaluated the system on TREC Clinical Trials data using accuracy, precision, recall, F1-score, and confusion matrices.

Wind Turbine Maintenance Advisor *GitHub*

NLP and Information Retrieval System

Python, Scikit-learn, TF-IDF, Cosine Similarity, JSON, LLMs

- Built an offline command-line advisor for wind-turbine maintenance, inspection, safety, and troubleshooting questions.
- Designed a structured JSON knowledge base covering 15 turbine components with rule-based intent and component detection.
- Implemented TF-IDF and cosine-similarity retrieval for evidence-backed answers from a technical maintenance manual.
- Integrated Groq/Llama as an optional rewriting layer while preserving complete offline functionality.

Brain Tumor Segmentation (U-Net) *GitHub*

Deep Learning (Medical Imaging)

Python, PyTorch, U-Net, CNN

- Built a U-Net model for brain tumor segmentation using the BraTS 2020 MRI dataset.
- Applied 5-fold cross-validation and evaluated segmentation performance using the Dice score.
- Compared the U-Net model with baseline segmentation approaches to assess performance improvements.